

Best LLMs for Extended Context Windows in 2026

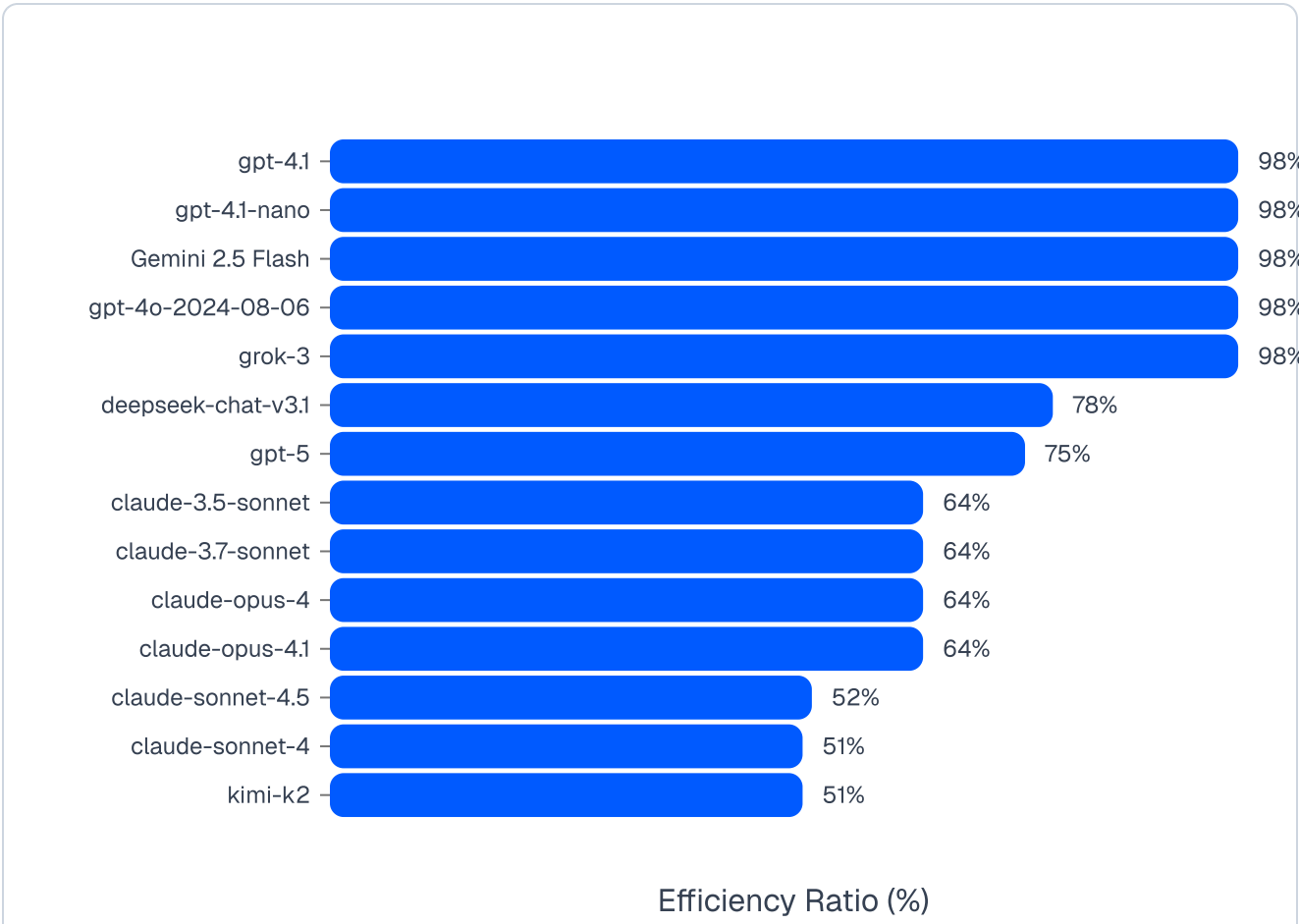


Cem Dilmegani updated on Jan 26, 2026

See our [ethical norms](#)

We ran a proprietary [32-message conversation test on 22 leading AI models](#) to see how much of their advertised context windows actually work. The conversation includes synthesis tasks that require recalling information from earlier messages, not just parroting the last thing said.

The chart below shows the efficiency ratios, indicating how much of each model’s advertised context window actually works in practice. See our full [methodology](#) for details on testing.



Key AI Models with Notable Context Window Capabilities

- **Magic LTM-2-Mini:** 100 million tokens with 1,000x efficiency improvement over traditional attention mechanisms. Requires a fraction of single H100 GPU vs. 638 H100s for comparable models. Purpose-built for software development. Limited production evidence as of January 2026, but represents the largest context window achieved to date.¹
- **Meta Llama 3.1:** Up to 128,000 tokens in some implementations with open-source flexibility but variable performance depending on hosting infrastructure²
- **Anthropic Claude 4 Sonnet:** 200,000 tokens standard, with 1M tokens available in beta for tier 4+ organizations (upgraded January 2026). Consistent performance with less than 5% accuracy degradation across the full context window³
- **OpenAI GPT-4 Turbo:** 128,000 tokens with reliable performance but noticeable slowdown and occasional inconsistencies when approaching maximum capacity⁴
- **Cohere Command-R+:** 128,000 tokens optimized for retrieval tasks with specialized architecture for maintaining context coherence⁵

Context Window Performance Comparison & Methodology

We systematically tested each model's ability to extract specific information from documents of varying lengths to find where performance declines and fails.

Most models break much earlier than advertised. A model claiming 200k tokens typically becomes unreliable around 130k, with sudden performance drops rather than gradual degradation.

Model	AI Memory Score	Advertised Context Window Limit (tokens)	Effective Context Window Limit (tokens)
Gemini-2.5-flash	67.5	1050k	~1025k
GPT-4.1	87.5	1050k	~1025k
GPT-4.1 Mini (nano)	87.5	1050k	~1025k
Claude Sonnet 4	82.5	1000k	~510k
Claude Sonnet 4.5	67.5	1000k	~510k

GPT-5	74.3	1000k	~408k
Claude Opus 4.1	85	200k	~130k
Claude Opus 4	85	200k	~130k
Claude 3.5 Sonnet	80	200k	~130k
Claude 3.7 Sonnet	67.5	200k	~130k

[Show More \(3\)](#)

Ranking Methodology

Rankings are based on the effective context window size, how well models retain, recall, and use information across sessions. [AI Memory](#) Score measures how consistently a model recalls information throughout a conversation, not just from the most recent messages. Higher scores mean the model maintains better awareness of earlier context.

Needle-in-a-Haystack testing

This test checks if models can find specific information buried in long documents. Difficulty increases sharply with document length and needle position.

- **Haystack:** Artificial documents with neutral, varied content at different lengths to prevent repetition patterns
- **Needle:** A distinct verification code inserted at specific locations, like CODE-A7B9C3D1E5F2
- **Task:** Find and extract the exact code when asked, “What is the verification code?”

Exponential ramp testing: Increases context exponentially to quickly find the approximate failure point instead of checking every length.

Binary search refinement: After failure, binary search pinpoints exactly where reliable performance ends.

Position sensitivity analysis: Tests whether needle position affects retrieval success at near-maximum reliable length, exposing “lost-in-the-middle” effects.

Evaluation: Models must respond with the exact format CODE-XXXX. Success is binary; either they find the correct code or they don’t. This eliminates subjective judgment.

AI Context Window Models and Pricing

Vendor	Representative model	Input (\$/1M)	Output (\$/1M)
Google	Gemini 2.5 Flash(Standard)	\$0.30	\$2.50
OpenAI	GPT-4o mini	\$0.15	\$0.60
Anthropic	Claude Sonnet 4(≤200k ctx)	\$3.00	\$15.00
Cohere	Command-R+ (Aug-2024 rate)	\$2.50	\$10.00
Mistral	Mistral Medium 3	\$0.40	\$2.00
DeepSeek	DeepSeek-V3(“deepseek-chat”)	\$0.27	\$1.10
xAI	Grok 4 Fast (reasoning)	\$0.20(<128k)	\$0.50
Meta	Llama 4 Scout(served via Together AI)	\$0.18	\$0.59
Google	Gemini 1.5 Pro	\$1.25 (≤128k) / \$2.50 (>128k)	\$5.00 (≤128k) / \$10.00 (>128k)

- All figures are per 1M tokens and shown in USD as of Sep 26, 2025

Below, you can see the most affordable models based on their effective context windows.

Detailed Model Profiles

1. OpenAI GPT-4.1 & GPT-4.1 Mini

The Mini variant delivers identical memory performance at significantly lower cost. Both handle 1M token contexts with consistent performance.⁶

Technical Strengths:

- Low hallucination rates when tested across full context range
- Handle interference questions without breaking focus on the primary task
- Extensive API ecosystem and third-party integrations

- Higher per-token pricing than open-source alternatives (\$2.50/\$10.00 per million tokens for standard, \$1.00/\$4.00 for Mini)
- API dependency creates vendor lock-in

Technical Characteristics:

- Mini variant offers identical performance at significantly reduced cost
- Robust handling of interference questions without performance degradation

Deployment Considerations: Suitable for applications requiring consistent accuracy across document types, particularly in regulated industries with compliance requirements

2. Meta Llama 4 Scout

Llama 4 features an absurd 10 million token context window the industry's largest. Uses a mixture of experts (MoE) architecture with 17B active parameters out of 109B total. ⁷

Technical Strengths:

- Complete customization and fine-tuning capabilities (open-source)
- No recurring API costs after deployment
- Native multimodal capabilities

Technical Limitations:

- Requires significant infrastructure investment for optimal performance
- Performance varies significantly based on the hosting configuration

Technical Characteristics:

- Mixture of experts (MoE) architecture with 17B active and 109B total parameters
- Native multimodal capabilities with an early fusion approach
- Variable hosting options from local deployment to cloud instances

DevStral achieved 61.6% on SWE-Bench Verified, beating both Gemini 2.5 Pro and GPT-4.1 at one-quarter the price. Purpose-built for coding with reinforcement learning optimization.⁸

Technical Strengths:

- State-of-the-art software engineering performance surpassing Gemini 2.5 Pro and GPT 4.1 at a quarter of the price
- Native GDPR compliance with EU data residency
- Purpose-built for **agentic coding** with reinforcement learning optimization
- On-premise deployment options for enhanced data privacy

Technical Characteristics:

- 128K token context window optimized for coding workflows
- Available through API at \$0.4/M input tokens and \$2/M output tokens
- Apache 2.0 license for community building and customization

Deployment Considerations: Suitable for European enterprises requiring GDPR compliance, software development teams, and organizations prioritizing data sovereignty

4. Anthropic Claude Sonnet 4 & Opus 4

Claude Sonnet 4 now offers 1M tokens in beta (upgraded from 200K standard) for organizations in usage tier 4 or with custom rate limits. Requests exceeding 200K are charged at 2x input and 1.5x output pricing.

Technical Strengths:

- Hybrid reasoning approach (fast default mode, extended thinking mode for complex problems)
- Advanced memory capabilities with local file access integration
- Tool use during extended thinking
- Context awareness tracks its own token budget throughout conversations

Technical Characteristics:

- A hybrid reasoning approach combining fast and deliberate responses

Deployment Considerations: Appropriate for applications in regulated environments where safety and explainability requirements outweigh maximum context length needs

5. Google Gemini 1.5 Pro & 2.5 Pro

Gemini offers the largest readily available context window at 2 million tokens with native multimodal processing across text, audio, images, and video.⁹

Technical Strengths:

- Native multimodal processing across multiple content formats
- Measured >99% retrieval accuracy in long-context benchmarks
- Context caching for cost optimization on repeated queries

Technical Limitations:

- Response latency increases significantly with very long contexts
- Computationally intensive requiring further latency optimizations

Technical Characteristics:

- Code execution capabilities for dynamic problem solving
- Multiple deployment options through Google Cloud Platform
- Near-perfect retrieval rates across most context ranges

Deployment Considerations: Suitable for applications requiring maximum context length, where processing time is less critical than comprehensive document analysis

6. OpenAI GPT-4 Turbo

The “old reliable” option with proven track record but smaller context window than newer alternatives.

Technical Strengths:

- Predictable behavior patterns across different use cases

Technical Limitations:

- Context window smaller than newer alternatives (128K vs 1M+ tokens)
- Performance degradation is observed when approaching maximum capacity

Technical Characteristics:

- 128K context window with consistent performance until near-maximum capacity
- 4K output token limit balances response quality with processing speed
- Well-optimized for common business use cases and integrations

Deployment Considerations: Suitable for standard business applications where proven reliability and ecosystem maturity are prioritized over maximum context length

7. xAI Grok-3 & Grok-4

Grok models integrating real-time web search with 2M token context and reinforcement learning-enhanced reasoning¹⁰.

Technical Strengths:

- Real-time information access with native web and X search capabilities
- Advanced reasoning capabilities refined through large scale reinforcement learning
- Native tool use and real-time search integration capabilities
- Specialized training on diverse internet content with current events understanding

Technical Limitations:

- Limited availability requiring X Premium+ subscription

Technical Characteristics:

- 256K context window available through API

- Strong performance across academic benchmarks including MMLU and AIME

Deployment Considerations: Suitable for applications requiring real-time information access, social media analysis, and current events tracking

8. DeepSeek-V3 & V3.1

DeepSeek models delivering cost-performance at \$0.48 per 1M tokens with hybrid thinking capabilities¹¹.

Technical Strengths:

- Open-source availability under MIT license
- 164K context window in V3.1 with hybrid thinking capabilities
- Requires only 2.788M H800 GPU hours for full training

Technical Limitations:

- Recommended deployment unit is relatively large, posing a burden for small teams

Technical Characteristics:

- 671B total parameters with 37B activated per token using MoE architecture
- Trained on 14.8 trillion tokens with focus on technical content
- 128K-164K context window with consistent performance across the full range

Deployment Considerations: Appropriate for software development, mathematical analysis, research applications, and cost-sensitive deployments requiring high technical capabilities

9. Cohere Command-R+

Command-R models are purpose-built for RAG workflows with specialized enterprise search and multilingual capabilities.

Technical Strengths:

- Multi-step tool use capabilities for complex business processes
- Advanced tool use with decision-making capabilities

Technical Characteristics:

- 128K context optimized for information synthesis
- Multilingual support across 10 key business languages
- Safety modes providing granular content control

Deployment Considerations: Suitable for enterprise knowledge management, customer support automation, and multilingual business operations requiring specialized RAG capabilities

FAQ

What is the Model Context Protocol (MCP) and why does it matter?



How do smaller models sometimes outperform larger ones in memory tasks?



What is the “lost in the middle” problem and how does it affect model selection?



Key Findings from Our Analysis:

- Context window size alone doesn’t determine performance quality
- Most models show degraded performance in the middle sections of long contexts
- Consistency across the full context range is often more valuable than maximum length
- Cost efficiency varies significantly between models and use cases

Further reading

- [AI Memory](#)
- [AI Agent Performance](#)

Reference Links

1. [Magic.dev](#)
2. [Meta Llama 3.1 \(70B\)](#)
3. [Claude Sonnet 4 now supports 1M tokens of context | Claude](#)
4. [What is the maximum context window for OpenAI's models?](#)
5. [Cohere's Command R+ Model | Cohere](#)
6. [Introducing GPT-4.1 in the API | OpenAI](#)
7. [The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation](#)
8. [Upgrading agentic coding capabilities with the new Devstral models | Mistral AI](#)
9. [Gemini 1.5 Pro 2M context window, code execution capabilities, and Gemma 2 are available today - Google Developers Blog](#)
10. [Grok 3 Beta — The Age of Reasoning Agents | xAI](#)
11. [DeepSeek V3 \(Dec\) - Intelligence, Performance & Price Analysis](#)



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Cem has been the principal analyst at AIMultiple since 2017. AIMultiple informs hundreds of thousands of businesses (as per similarWeb) including 55% of Fortune 500 every month.

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Throughout his career, Cem served as a tech consultant, tech buyer and tech entrepreneur. He advised enterprises on their technology decisions at McKinsey & Company and Altman Solon for more than a decade. He also published a McKinsey report on digitalization.

He led technology strategy and procurement of a telco while reporting to the CEO. He has also led commercial growth of deep tech company Hypatos that reached a 7 digit annual recurring revenue and a 9 digit valuation from 0 within 2 years. Cem's work in Hypatos was covered by leading technology publications like TechCrunch and Business Insider.

Cem regularly speaks at international technology conferences. He graduated from Bogazici University as a computer engineer and holds an MBA from Columbia Business School.

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